PROJECT REPORT

Electric Vehicle Market Performance & Insights

Data Analytics with Tableau – Capstone Project

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Team ID: NA

November 2026

Dashboard:
https://public.tableau.com/app/profile/parvi.sharma/viz/ElectricVehicleMarketPerformanceandInsights/EVMarketDashboard

Story:
https://public.tableau.com/app/profile/parvi.sharma/viz/ElectricVehicleMarketPerformanceandInsights/Story1

GitHub: [Paste GitHub repository link]

Demo video: [Paste Google Drive demo video link]

1. INTRODUCTION

1.1 Project Overview

Electric vehicles are one of the fastest-growing parts of the automotive industry. Manufacturers, consultants and governments need clear information about which vehicles sell, where they sell, which battery technologies are used and how much CO_2 they save.

In this project I analysed 100,000 EV records (ev_market_test.csv, 37 attributes, model years 2018–2026) in Tableau. I built worksheets, an EV Market Dashboard and a Tableau Story, and published them on Tableau Public.

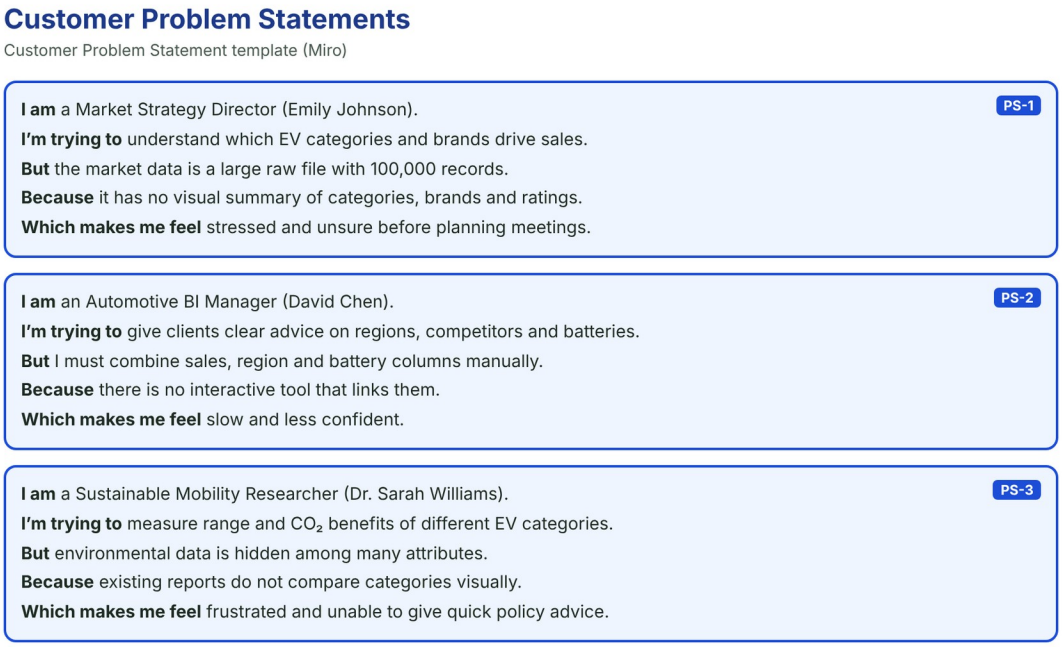
1.2 Purpose

The dashboard answers the questions of three users:

- **Emily Johnson (EV Market Strategy Director):** wants to see which vehicle categories bring the most annual sales, which manufacturers lead, how satisfied customers are in each segment, which batteries are being adopted and how driving range differs across categories.
- **David Chen (Automotive BI Manager):** advises clients entering the EV market and needs manufacturer market share, sales performance, battery technology mix, regional/manufacturing analysis and customer ratings.
- **Dr. Sarah Williams (Sustainable Mobility Research Analyst):** studies battery technologies, driving range, global manufacturing distribution, customer satisfaction and environmental impact to advise policymakers.

2. IDEATION PHASE

2.1 Problem Statement



2.2 Empathy Map Canvas

Empathy Map – Who uses the EV Market Dashboard?

Users: Emily Johnson (Strategy Director) · David Chen (BI Manager) · Dr. Sarah Williams (Research Analyst)

SAYS
- "Show me which vehicle types sell the most."
- "Which brands should my clients watch?"
- "How much CO_2 do electric buses really save?"
- "I need answers I can present quickly."

THINKS
- 100,000 rows of raw data is hard to read
- Is demand different between regions?
- Which battery type will win in the long run?
- Are long-range vehicles worth the price?

DOES
- Opens CSV files and filters them manually
- Builds one-off charts for every meeting
- Searches reports on batteries and emissions
- Compares manufacturers by sales and ratings

FEELS
- Confused by too many columns
- Worried about missing important trends
- Under time pressure before reviews
- Relieved when data is clear and visual

PAINS
- No single view of sales, regions, brands and batteries
- Manual work is slow and repeated
- Hard to compare categories side by side
- Environmental impact is not visible

GAINS
- One Tableau dashboard with 6 clear charts
- Instant comparison of categories, regions and brands
- Battery mix and CO_2 savings shown visually
- Public link and story to share with stakeholders

2.3 Brainstorming

Ideas were grouped and ranked by feasibility and importance. A Tableau dashboard for sales, regions and manufacturers, sustainability views (battery, range, CO_2) and a published story were selected; a price prediction model was not selected.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Customer Journey Map – EV Dashboard User

Persona: Emily Johnson (also representing David Chen and Dr. Sarah Williams)

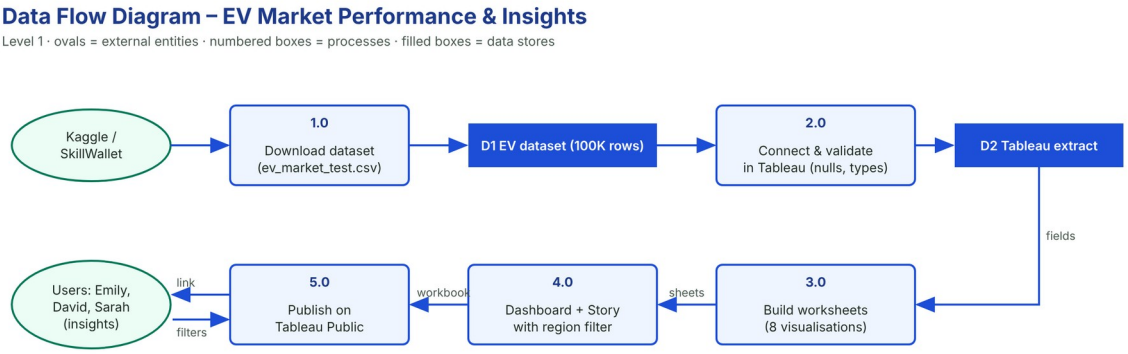
	1. Question	2. Explore	3. Act
Actions	Gets a question on sales, brands or emissions	Opens the dashboard and uses the region filter	Shares the story and recommends actions
Touchpoints	Meeting / email, raw dataset	Category, region, range, CO_2, battery, top-10 charts	Tableau Story, Tableau Public link
Thoughts	"Where is the answer in 100,000 rows?"	"Two-wheelers sell most; buses save most CO_2."	"I can explain this in two minutes."
Feelings	😟 Stressed	🙂 Interested	😊 Confident
Ownership	Business user	Analyst (Parvi Sharma)	Business user
Opportunities	Keep data in one place	Add more filters and tooltips	Add forecasting in future

3.2 Solution Requirement

FR	Requirement	Description
FR-1	Data access	Connect ev_market_test.csv to Tableau as an extract
FR-2	Data processing	Check types, nulls, duplicates
FR-3	Analysis	SUM, AVG, COUNT, Top-10 rankings
FR-4	Outputs	8 worksheets, dashboard,

		story
FR-5	Interaction	Region filter, tooltips, sorting
FR-6	Reporting	Tableau Public link

3.3 Data Flow Diagram



3.4 Technology Stack

Tableau Desktop and Tableau Public (visualisation and publishing), CSV + Tableau extract (data), Tableau calculations (aggregation), GitHub (files and documentation), Google Drive (demo video).

4. PROJECT DESIGN

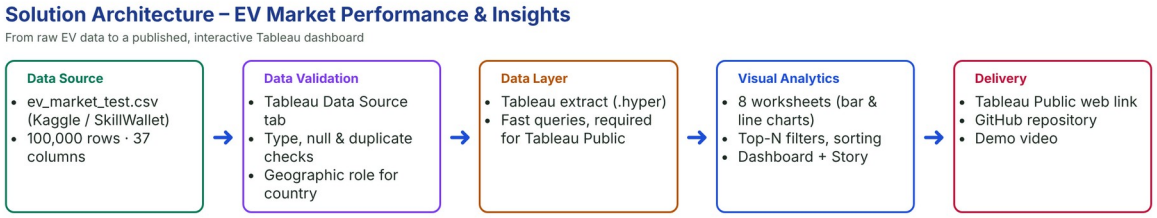
4.1 Problem Solution Fit



4.2 Proposed Solution

Download the data → check it in Tableau → create an extract → build worksheets → design a dashboard and story → publish on Tableau Public → test and document.

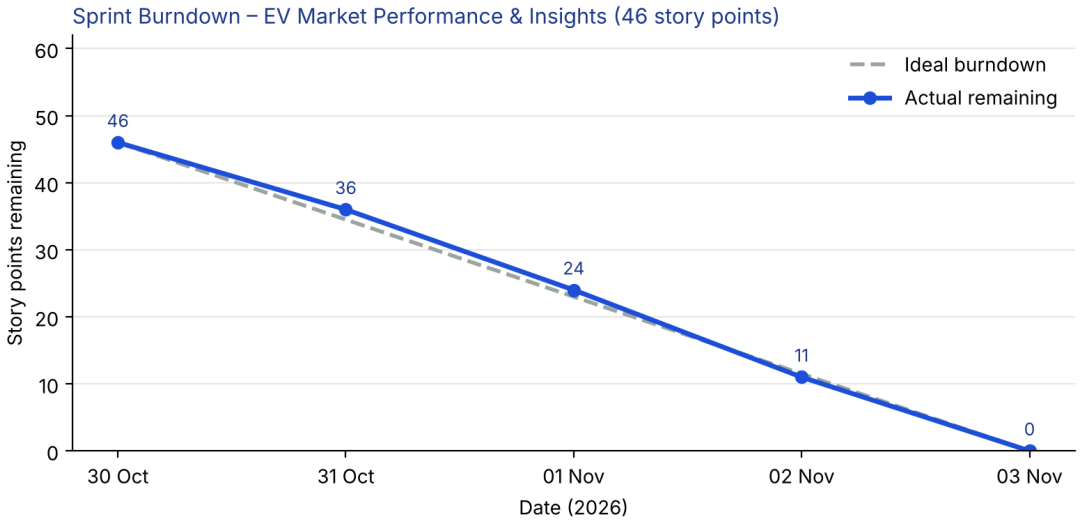
4.3 Solution Architecture



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

19 user stories (46 story points) were completed in 4 one-day sprints between 30 October and 3 November 2026; average velocity 11.5 points per sprint.



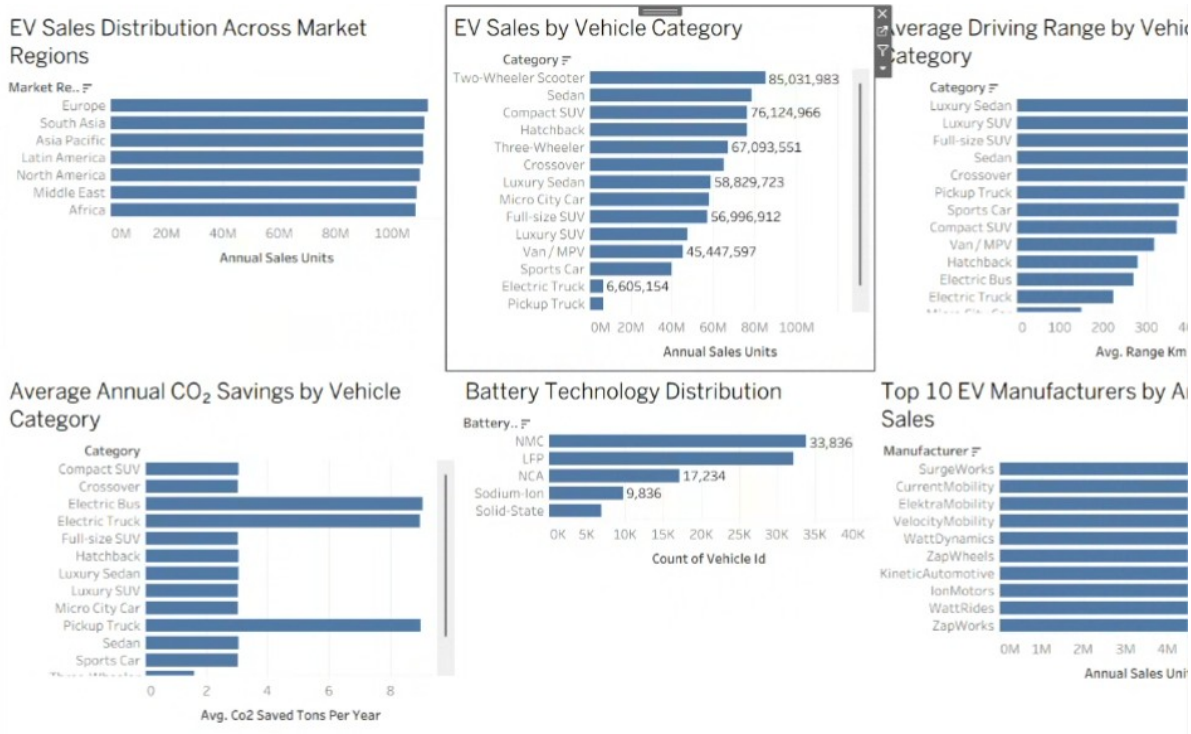
6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Parameter	Values
Data Rendered	100,000 rows × 37 columns in a Tableau extract
Data Checks	0 duplicates; nulls in 5 columns ignored by SUM/AVG
Filters	Market Region; Top-10 on Manufacturer
Aggregations	SUM([Annual Sales Units]); AVG([Range km]); AVG([CO2 Saved Tons per Year]); COUNT([Vehicle Id]); AVG([Customer Rating])
Dashboard	6 charts; 8 worksheets in total
Story	2 story points

7. RESULTS

7.1 Output Screenshots



EV Market Dashboard

1. EV Sales by Vehicle Category

Question: Which vehicle categories generate the highest annual EV sales? (Scenario 1, 2)



Insight: Two-Wheeler Scooters sell the most (85.0 M units), followed by Sedans (78.3 M), Compact SUVs (76.1 M) and Hatchbacks (76.1 M). Heavy vehicles sell the least – Electric Trucks 6.6 M, Pickup Trucks 6.5 M and Electric Buses 5.3 M.

2. EV Sales Distribution Across Market Regions

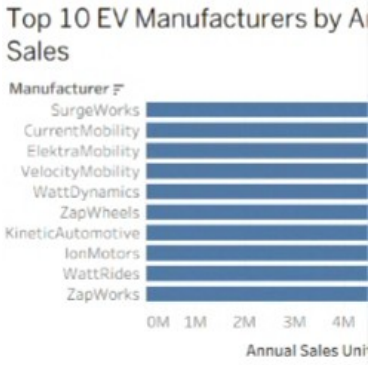
Question: How are EV sales spread across the world's market regions? (Scenario 2)



Insight: Sales are very evenly spread: Europe leads with 112.5 M units, followed by South Asia (111.3 M) and Asia Pacific (111.1 M), while Africa is lowest at 108.3 M. Every region holds about 14% of the market, so no single region dominates.

3. Top 10 EV Manufacturers by Annual Sales

Question: Which manufacturers lead the EV market by annual sales? (Scenario 1, 2)



Insight: SurgeWorks leads with 4.88 M units, followed by CurrentMobility (4.78 M) and ElektraMobility (4.75 M). The top 10 brands together hold only about 6% of total sales, showing a highly fragmented, competitive market.

4. Top 10 Manufacturers by Customer Rating

Question: Which manufacturers have the most satisfied customers? (Scenario 1, 2)

INSERT TABLEAU SCREENSHOT

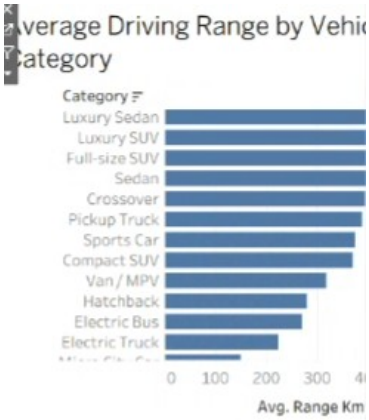
Top 10 Manufacturers by Rating

(replace this box with your screenshot)

Insight: ElektraDynamics has the highest average customer rating (4.17 / 5), followed by ChargeWorks (4.15) and WattMotors (4.14). The overall average is 4.08, so the best brands are only slightly ahead – satisfaction is high across the market.

5. Average Driving Range by Vehicle Category

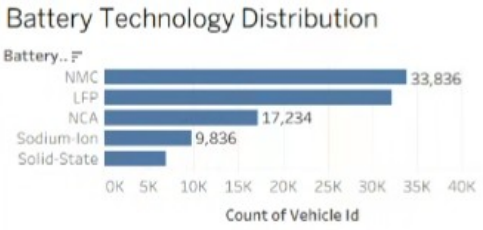
Question: How does driving range compare across vehicle categories? (Scenario 1, 3)



Insight: Luxury Sedans have the longest average range (549 km), then Luxury SUVs (525 km) and Full-size SUVs (476 km). Two-Wheeler Scooters (105 km) and Three-Wheelers (130 km) have the shortest range, matching short urban trips.

6. Battery Technology Distribution

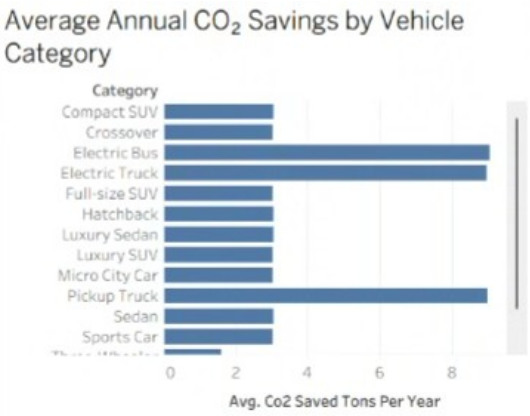
Question: Which battery technologies are most widely used? (Scenario 1, 2, 3)



Insight: NMC (33,836 vehicles) and LFP (32,101) are used in about two-thirds of all EVs, followed by NCA (17,234), Sodium-Ion (9,836) and Solid-State (6,993). Solid-State is still an emerging technology.

7. Average Annual CO_2 Savings by Vehicle Category

Question: Which vehicle categories save the most CO_2 every year? (Scenario 3)



Insight: Electric Buses (9.05 t/yr), Pickup Trucks (9.01) and Electric Trucks (8.96) save about three times more CO_2 per vehicle than passenger cars (≈3.0 t/yr), while two- and three-wheelers save about 1.6 t/yr. Electrifying heavy vehicles gives the biggest environmental benefit.

8. EV Sales by Model Year

Question: How have annual EV sales changed across model years 2018–2026? (Scenario 1, 3)



Insight: Sales stay in a steady band of 82.6 M–88.1 M units per model year. The 2021 models sold the most (88.1 M) and the 2025 models the least (82.6 M), so demand is stable rather than falling.

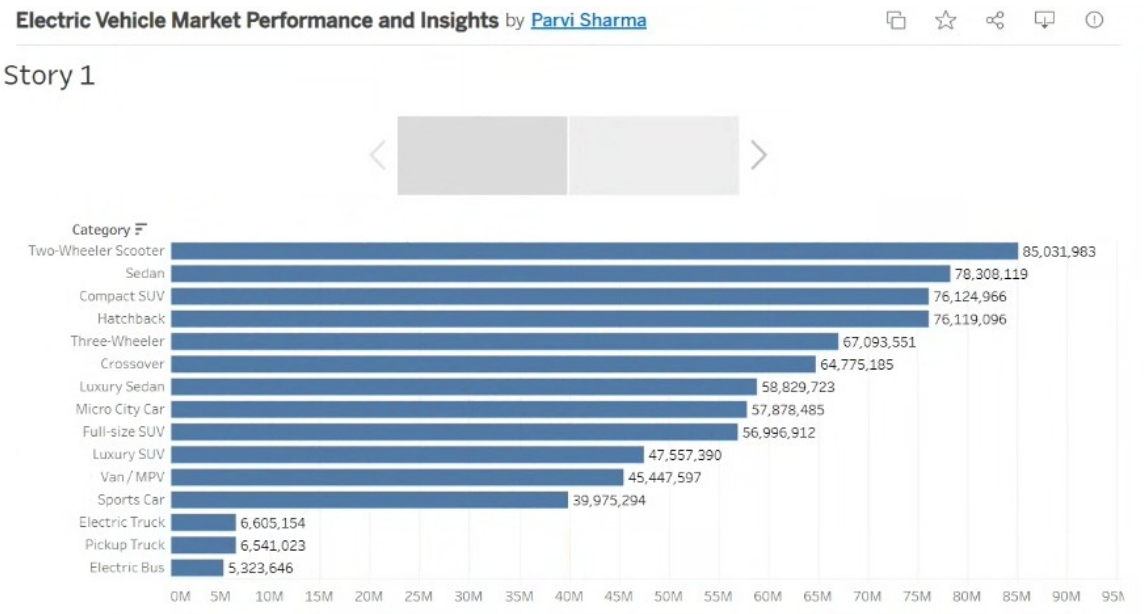


Tableau Story

8. ADVANTAGES & DISADVANTAGES

Advantages

- Answers all three scenarios in one place.
- Easy to read – sorted bar charts with clear titles.
- Free and shareable through Tableau Public.
- Includes environmental impact (CO_2 savings).

Disadvantages

- Static data file, no live refresh.
- Only a region filter on the dashboard.
- No forecasting yet.

9. CONCLUSION

The dashboard shows that affordable urban EVs (two-wheeler scooters, sedans, compact SUVs) drive sales, demand is balanced across all seven regions, no manufacturer dominates (top 10 ≈ 6% of sales), NMC and LFP batteries power about two-thirds of vehicles, luxury vehicles offer the longest range and heavy electric vehicles save the most CO_2. These insights help Emily plan products, David advise clients and Sarah support clean-mobility policy.

10. FUTURE SCOPE

- Add more filters and drill-down.
- Add sales forecasting.
- Connect to live data with scheduled refresh.
- Add price and charging analysis.

11. APPENDIX

Item	Link
Dataset	https://www.kaggle.com/datasets/tharishreddy22/electric-vehicle-market-classification-dataset
Dashboard	https://public.tableau.com/app/profile/parvi.sharma/viz/ElectricVehicleMarketPerformanceandInsights/EVMarketDashboard
Story	https://public.tableau.com/app/profile/parvi.sharma/viz/ElectricVehicleMarketPerformanceandInsights/Story1
GitHub	[Paste GitHub repository link]
Demo video	[Paste Google Drive demo video link]

Data checks performed:

- Used the ev_market_test.csv file from the SkillWallet / Kaggle dataset: 100,000 rows × 37 columns.
- Checked the data in Tableau's Data Source tab: correct data types for 37 fields; Manufacturing Country set as a geographic field.

• Duplicate check: 0 duplicate rows.

• Null check: a few missing values in Annual Sales Units (989), Customer Rating (1,931), Resale Value (3,057), Battery Cycle Life (2,028) and Screen Size (1,506). Tableau's SUM and AVG ignore nulls, so no rows had to be removed.

• Created a Tableau extract (.hyper) so the dashboard loads fast and can be published to Tableau Public.